

## JAGDALPUR, India

WMO: 430410

Lat: 19.083N

Lon: 82.033E

Elev: 553

StdP: 94.86

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 10.3       | 11.5 | 5.6                           | 6.0 | 23.2 | 6.9 | 6.6 | 19.9 | 3.1                   | 22.4 | 2.5 | 23.9 | 0.0                   | 180  | 0.257 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 12.1                   | 39.3            | 22.9 | 38.1 | 23.0 | 36.9 | 22.9 | 26.6                | 33.0 | 26.0 | 31.6 | 25.6 | 30.6 | 1.4                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.1                            | 21.7 | 28.8 | 24.7 | 21.1 | 28.0 | 24.4 | 20.7 | 27.6 | 87.3          | 33.3 | 84.5 | 31.8 | 82.6 | 30.6 | 34.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 3.2               | 2.5  | 2.3 | 7.7                        | 41.1 | 1.2                | 1.4 | 6.8                                                | 42.1 | 6.1        | 42.9 | 5.4        | 43.6 | 4.5        | 44.6 |
|                   |      |     | 6.9                        | 28.3 | 1.2                | 1.8 | 6.0                                                | 29.6 | 5.2        | 30.6 | 4.5        | 31.6 | 3.6        | 32.9 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 25.4        | 20.3  | 23.3  | 27.1  | 29.9  | 30.9  | 28.3  | 25.7  | 25.5  | 26.1  | 25.3  | 22.5  | 20.1  | (6)  |
|                                                                        | DBStd       | 3.78        | 1.82  | 1.94  | 1.93  | 2.01  | 2.09  | 2.87  | 1.55  | 1.36  | 1.20  | 1.42  | 1.93  | 1.90  | (7)  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD18.3     | 9           | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5     | (9)  |
|                                                                        | CDD10.0     | 5631        | 320   | 373   | 530   | 598   | 647   | 549   | 486   | 482   | 484   | 474   | 375   | 314   | (10) |
|                                                                        | CDD18.3     | 2599        | 66    | 141   | 272   | 348   | 388   | 299   | 228   | 223   | 234   | 216   | 126   | 60    | (11) |
|                                                                        | CDH23.3     | 26586       | 717   | 1486  | 3216  | 4622  | 5329  | 3294  | 1578  | 1461  | 1724  | 1641  | 939   | 579   | (12) |
|                                                                        | CDH26.7     | 11865       | 144   | 631   | 1752  | 2698  | 3154  | 1552  | 358   | 296   | 457   | 512   | 222   | 88    | (13) |
| (14) Wind                                                              | WSAvg       | 0.6         | 0.4   | 0.5   | 0.6   | 0.8   | 0.9   | 0.9   | 0.9   | 0.7   | 0.6   | 0.5   | 0.4   | 0.3   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 1473        | 9     | 10    | 19    | 54    | 80    | 250   | 340   | 359   | 225   | 91    | 21    | 8     | (15) |
|                                                                        | PrecMax     | 2262        | 64    | 69    | 136   | 142   | 342   | 544   | 615   | 757   | 490   | 299   | 125   | 122   | (16) |
|                                                                        | PrecMin     | 921         | 0     | 0     | 0     | 0     | 2     | 49    | 20    | 117   | 75    | 0     | 0     | 0     | (17) |
|                                                                        | PrecStd     | 257         | 15    | 16    | 26    | 39    | 53    | 115   | 109   | 120   | 94    | 79    | 31    | 22    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 31.2  | 34.3  | 37.7  | 40.0  | 41.3  | 40.5  | 32.4  | 31.2  | 32.1  | 32.1  | 30.8  | 29.8  | (19) |
|                                                                        |             | MCWB        | 19.6  | 19.9  | 20.7  | 22.1  | 22.7  | 23.1  | 25.3  | 25.3  | 24.9  | 23.0  | 21.6  | 19.3  | (20) |
|                                                                        | 2%          | DB          | 29.2  | 32.9  | 36.4  | 38.7  | 39.9  | 38.0  | 30.9  | 30.3  | 31.1  | 31.1  | 29.8  | 28.6  | (21) |
|                                                                        |             | MCWB        | 18.4  | 19.3  | 20.7  | 22.1  | 23.0  | 24.3  | 25.2  | 24.9  | 24.7  | 23.0  | 21.2  | 19.1  | (22) |
|                                                                        | 5%          | DB          | 28.2  | 31.7  | 35.4  | 37.7  | 38.6  | 35.7  | 29.8  | 29.5  | 30.2  | 30.4  | 29.0  | 27.7  | (23) |
|                                                                        |             | MCWB        | 18.2  | 19.1  | 20.8  | 22.3  | 23.5  | 24.8  | 24.9  | 24.7  | 24.6  | 22.9  | 20.8  | 18.8  | (24) |
|                                                                        | 10%         | DB          | 27.2  | 30.5  | 34.2  | 36.6  | 37.1  | 33.6  | 28.7  | 28.4  | 29.4  | 29.5  | 28.1  | 26.7  | (25) |
|                                                                        |             | MCWB        | 17.8  | 18.9  | 20.6  | 22.3  | 23.9  | 24.8  | 24.7  | 24.5  | 24.6  | 22.8  | 20.5  | 18.4  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 21.9  | 22.2  | 24.7  | 25.9  | 27.6  | 27.3  | 26.4  | 26.1  | 26.2  | 25.7  | 24.8  | 22.4  | (27) |
|                                                                        |             | MCDB        | 27.5  | 29.2  | 32.3  | 34.4  | 35.9  | 33.8  | 29.9  | 29.6  | 30.1  | 29.3  | 28.8  | 27.7  | (28) |
|                                                                        | 2%          | WB          | 20.8  | 21.4  | 23.5  | 25.1  | 26.7  | 26.5  | 25.7  | 25.5  | 25.6  | 25.0  | 23.7  | 21.3  | (29) |
|                                                                        |             | MCDB        | 26.3  | 28.7  | 31.7  | 33.4  | 34.6  | 32.5  | 29.2  | 28.7  | 29.2  | 28.8  | 27.7  | 25.9  | (30) |
|                                                                        | 5%          | WB          | 19.7  | 20.7  | 22.7  | 24.5  | 26.0  | 25.9  | 25.3  | 25.1  | 25.2  | 24.5  | 23.0  | 20.3  | (31) |
|                                                                        |             | MCDB        | 25.2  | 28.1  | 30.8  | 32.9  | 33.4  | 31.5  | 28.4  | 28.1  | 28.6  | 28.1  | 26.5  | 25.2  | (32) |
|                                                                        | 10%         | WB          | 18.9  | 20.1  | 22.0  | 23.9  | 25.3  | 25.5  | 24.9  | 24.7  | 24.8  | 24.1  | 22.3  | 19.5  | (33) |
|                                                                        |             | MCDB        | 25.1  | 27.5  | 29.9  | 32.3  | 32.6  | 30.5  | 27.8  | 27.3  | 28.0  | 27.4  | 25.6  | 24.6  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 14.9  | 14.7  | 14.3  | 13.3  | 12.1  | 7.8   | 5.0   | 5.0   | 6.3   | 9.0   | 12.3  | 14.4  | (35) |
|                                                                        |             | MCDBR       | 15.8  | 16.1  | 15.6  | 14.4  | 13.9  | 11.6  | 7.2   | 7.1   | 7.9   | 10.3  | 13.0  | 15.3  | (36) |
|                                                                        | 5% WB       | MCWBR       | 6.6   | 5.5   | 4.8   | 4.0   | 3.6   | 3.2   | 2.7   | 2.6   | 2.7   | 3.6   | 5.3   | 6.9   | (37) |
|                                                                        |             | MCWBR       | 13.0  | 13.3  | 13.4  | 12.9  | 11.7  | 8.8   | 6.4   | 6.0   | 6.8   | 7.7   | 8.9   | 12.2  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.410 | 0.434 | 0.539 | 0.552 | 0.579 | 0.619 | 0.553 | 0.527 | 0.526 | 0.495 | 0.439 | 0.396 | (40) |
|                                                                        |             | taud        | 2.205 | 2.141 | 1.878 | 1.909 | 1.859 | 1.799 | 1.981 | 2.066 | 2.061 | 2.108 | 2.191 | 2.263 | (41) |
|                                                                        | Ebn at Noon |             | 862   | 862   | 784   | 777   | 748   | 712   | 762   | 786   | 785   | 796   | 826   | 864   | (42) |
|                                                                        |             | Edh at Noon | 135   | 152   | 204   | 200   | 208   | 219   | 183   | 169   | 168   | 155   | 136   | 124   | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 4.79  | 5.59  | 6.10  | 6.59  | 6.52  | 4.67  | 3.50  | 3.54  | 4.31  | 4.89  | 4.73  | 4.48  | (44) |
|                                                                        | RadStd      |             | 0.18  | 0.21  | 0.32  | 0.27  | 0.19  | 0.55  | 0.54  | 0.40  | 0.37  | 0.42  | 0.31  | 0.28  | (45) |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page